

DIGITAL POWER SUPPLY DESIGN

Using A Microcontroller



The author, **SANTOSH GURRAL**, is an Infineon system hardware engineer with over 13 years of embedded industry experience. He specialises in designing and developing embedded products, with expertise in high-speed board design, digital, analogue, and power supply circuits. Proficient in PCB layout, simulation tools, and EMI/EMC testing.

As power demands rise, digital power supplies provide a forward-thinking solution, optimising efficiency and reducing component complexity through digital control.

A typical DC-DC power supply, such as a buck power supply, comprises an input supply, switching network, and passive elements (L and C) for filtering, feedback, compensation network, pulse width modulation (PWM), reference voltage, and output power.

A digital power supply replaces the analogue compensation network with an analogue-to-digital converter (ADC), compensation algorithm, and digital PWM implemented inside the microcontroller.

They enable systems design with multiple and variable outputs while minimising real estate requirements on circuit boards. A single microcontroller (MCU) can implement multiple power rails, significantly enhancing design efficiency.

Furthermore, digital power supplies deliver higher efficiency across the entire operational range, offering the capability to combine multiple control loops or functions into a single device. The flexibility provided by software-programmable solutions allows for adjustments in control loop performance through software modifications, ensuring adaptability to changing requirements.

Additionally, these systems support advanced features such as auto-tuning, self-testing, power-up sequence setups, communication across other channels, and even failure prediction, making them versatile and future-ready.

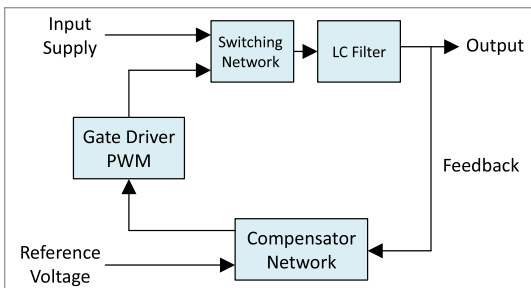


Fig. 1: DC-DC power supply system (Credit: www.infineon.com)

Advantages of digital power supply

Digital power supplies present numerous benefits, primarily by eliminating components and reducing

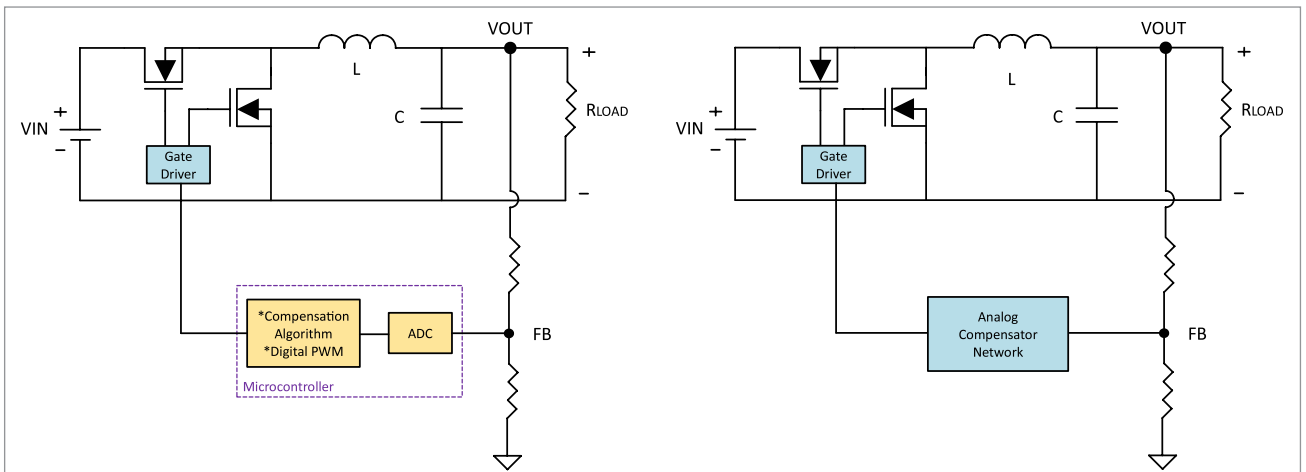


Fig. 2: Digital power supply system with microcontroller (Credit: www.infineon.com)